



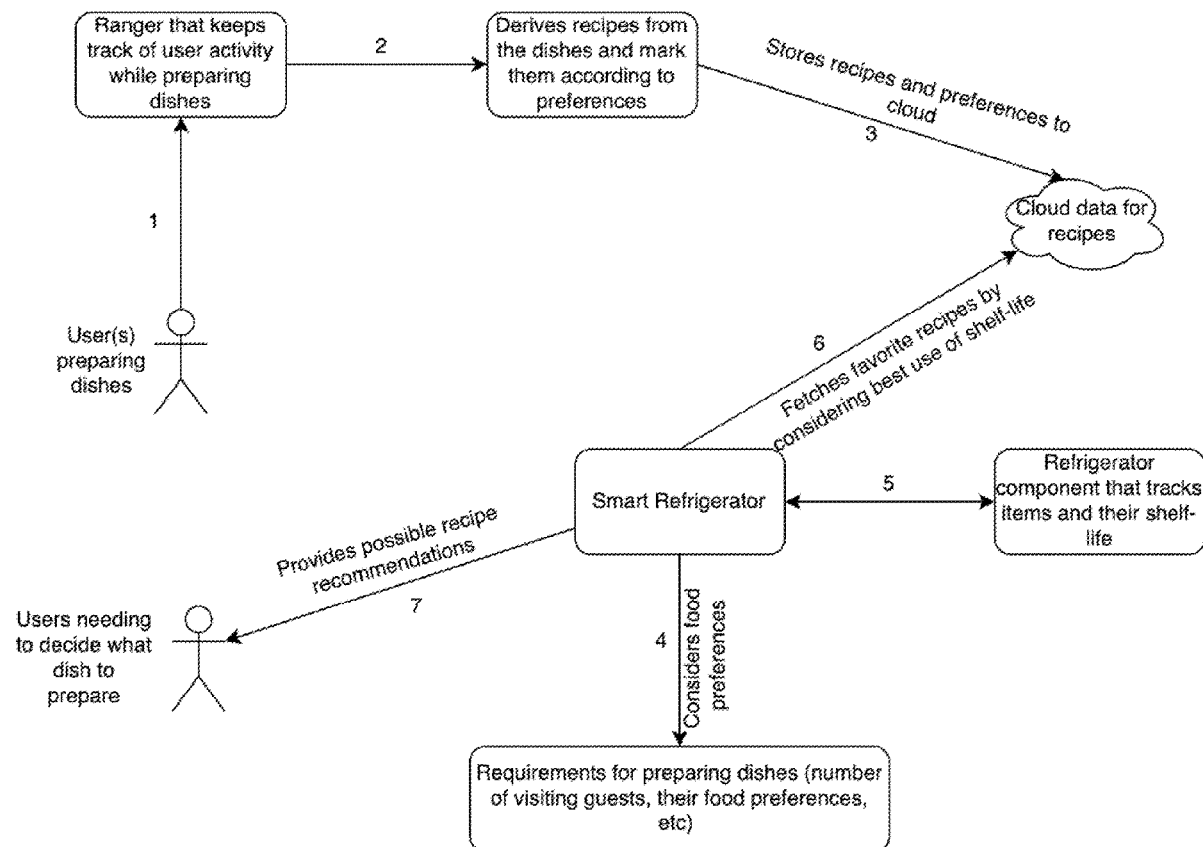
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(19) **United States**(12) **Patent Application Publication**
Perumalla et al.(10) **Pub. No.: US 2022/0263680 A1**(43) **Pub. Date: Aug. 18, 2022**(54) **SMART REFRIGERATOR FOR SMART
CONSUMPTION OF FOOD***G16H 20/60* (2006.01)*G16H 50/30* (2006.01)*G06K 9/62* (2006.01)(71) Applicant: **INTERNATIONAL BUSINESS
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ABSTRACT(21) Appl. No.: **17/248,945**(22) Filed: **Feb. 15, 2021****Publication Classification**(51) **Int. Cl.***H04L 12/28* (2006.01)*G16H 10/60* (2006.01)

The exemplary embodiments disclose a method, a computer program product, and a computer system for managing a consumption of food. The exemplary embodiments may include collecting data of a user and of one or more food items, extracting one or more features from the collected data, and determining a food consumption recommendation based on the extracted one or more features and one or more models.

Smart Ranger that intelligently derives favorite recipes from live cooking

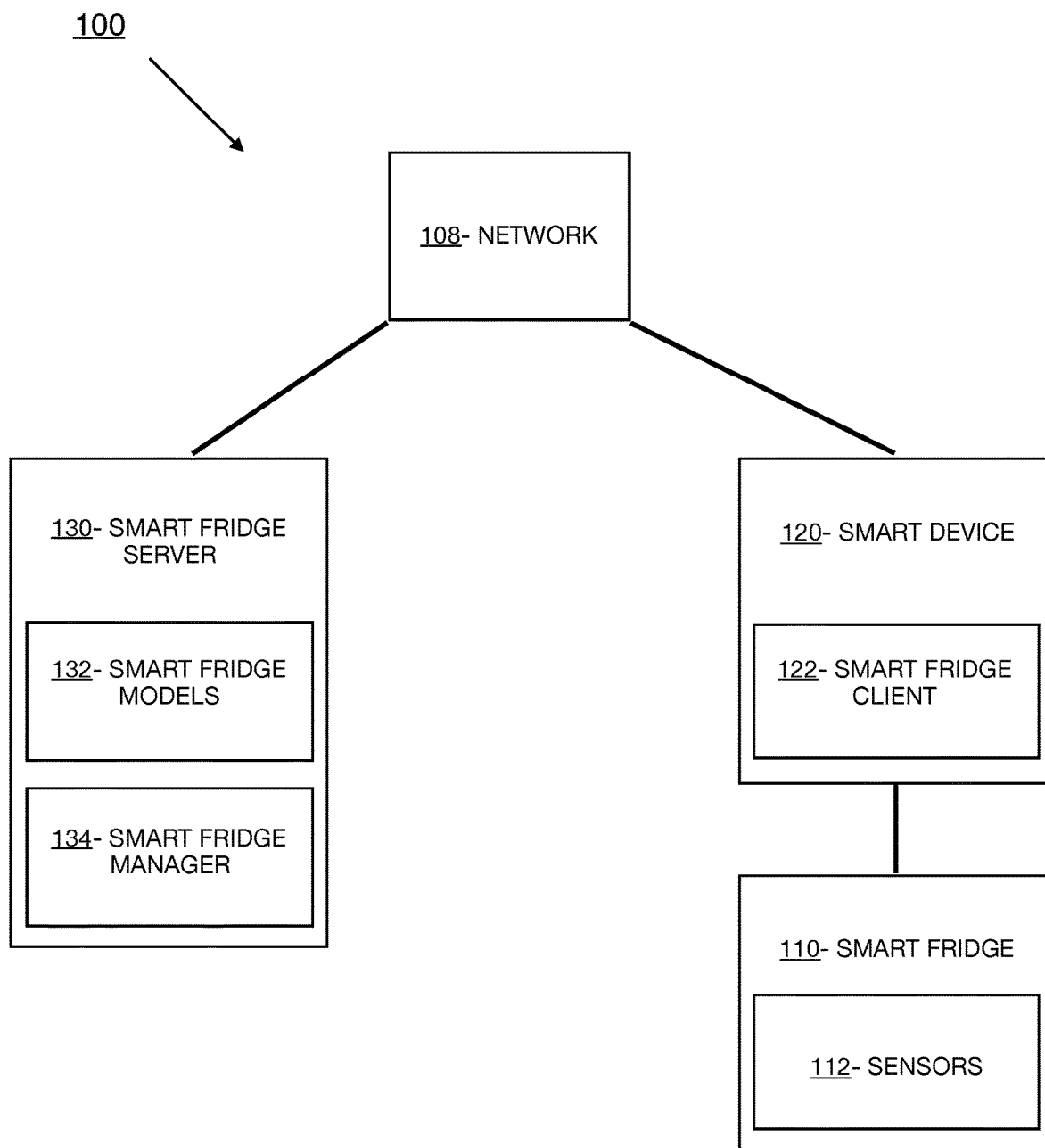


FIG. 1

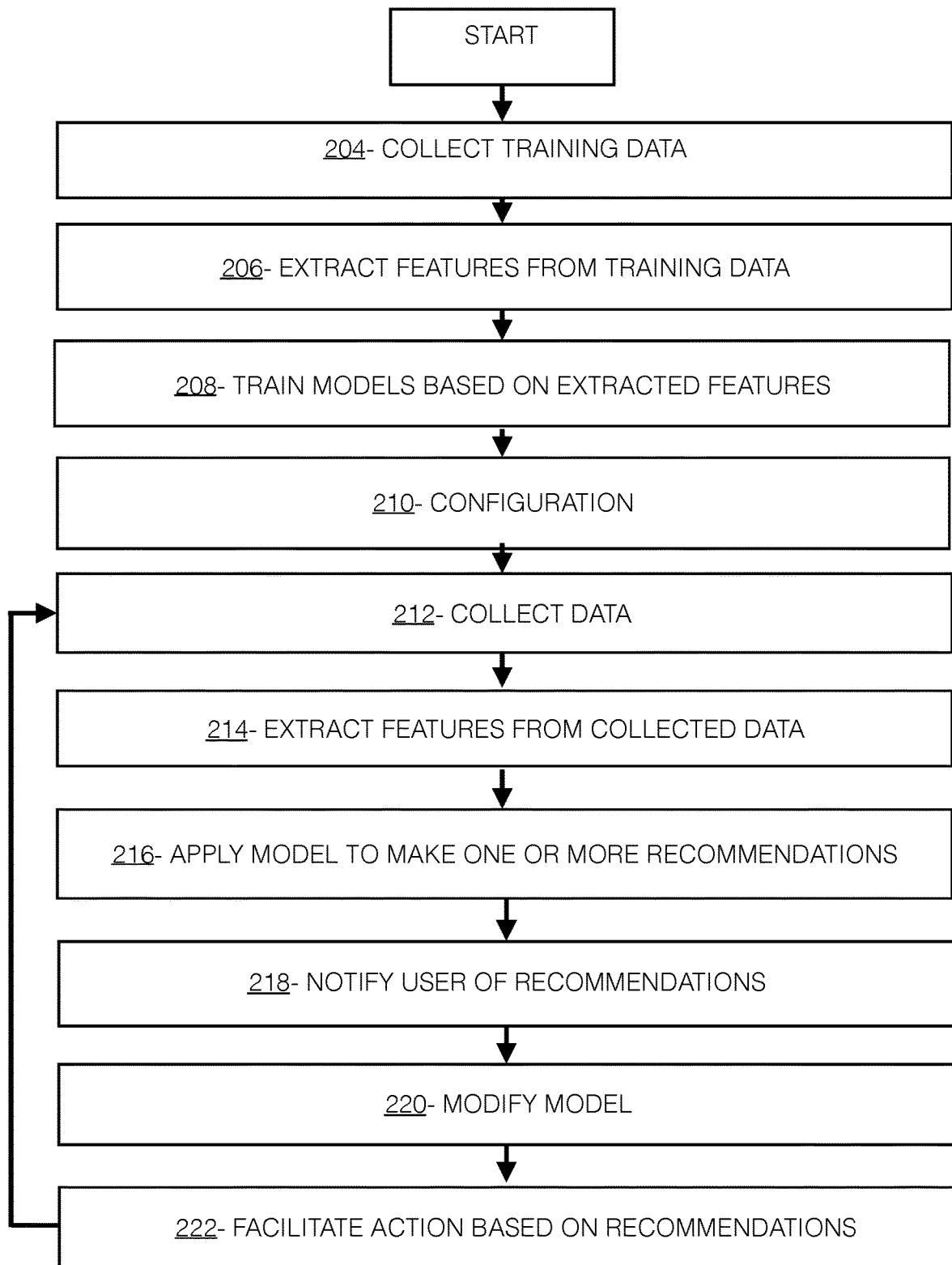


FIG. 2

Smart Ranger that intelligently derives favorite recipes from live cooking

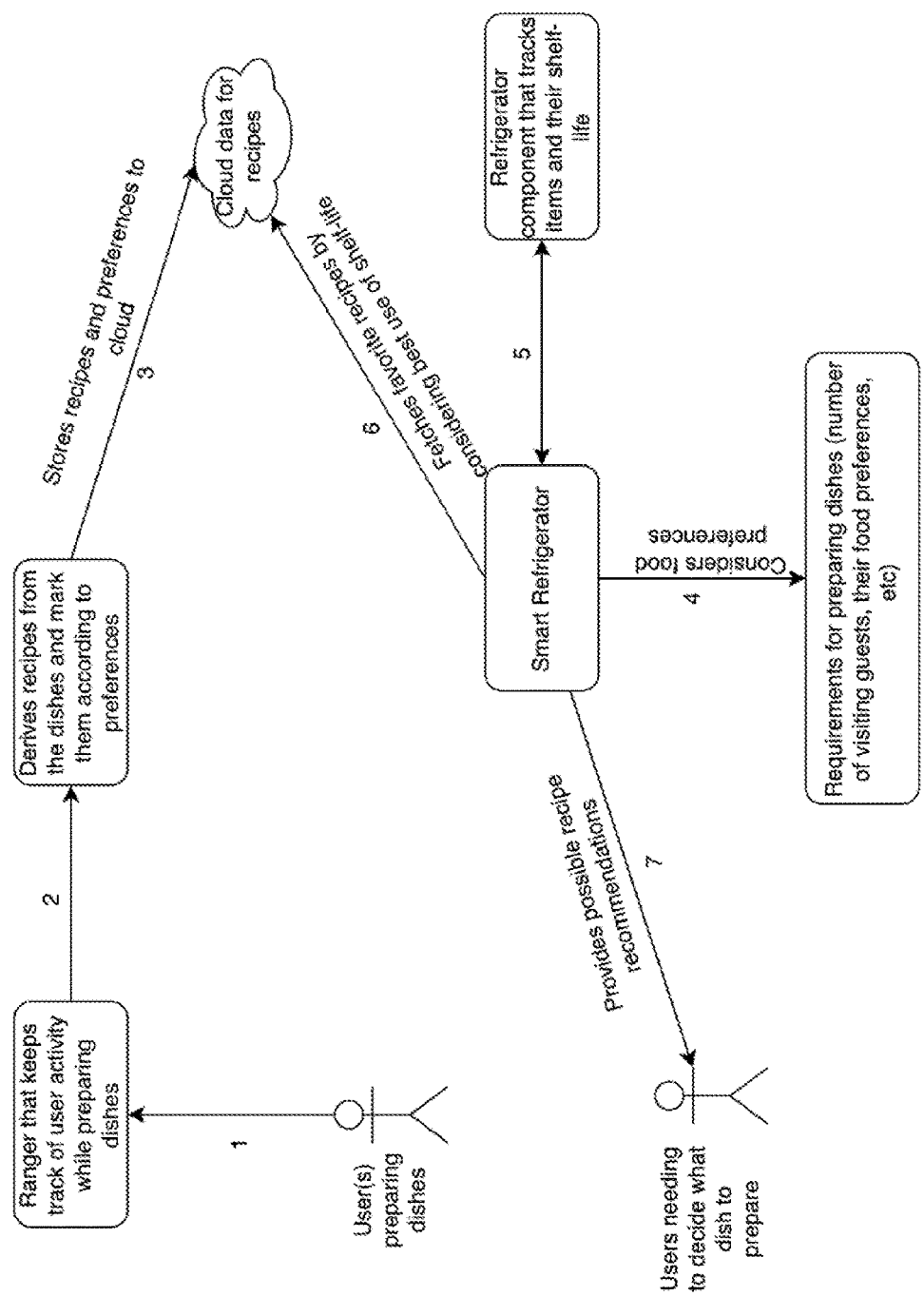


FIG. 3

Smart Refrigerator for identifying and scheduling expiring items give away

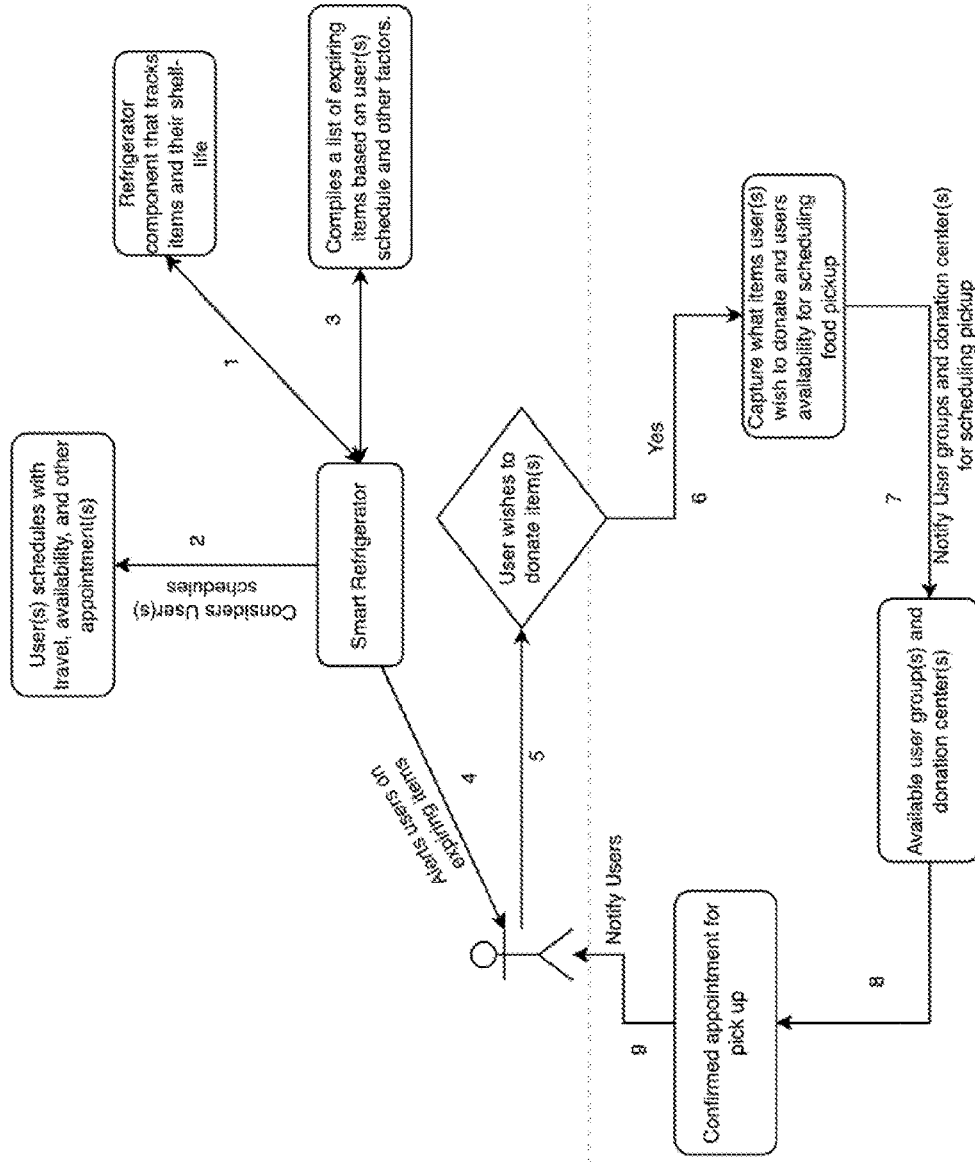


FIG. 4

Smart Refrigerator for physician consultation and recommending customized diet options

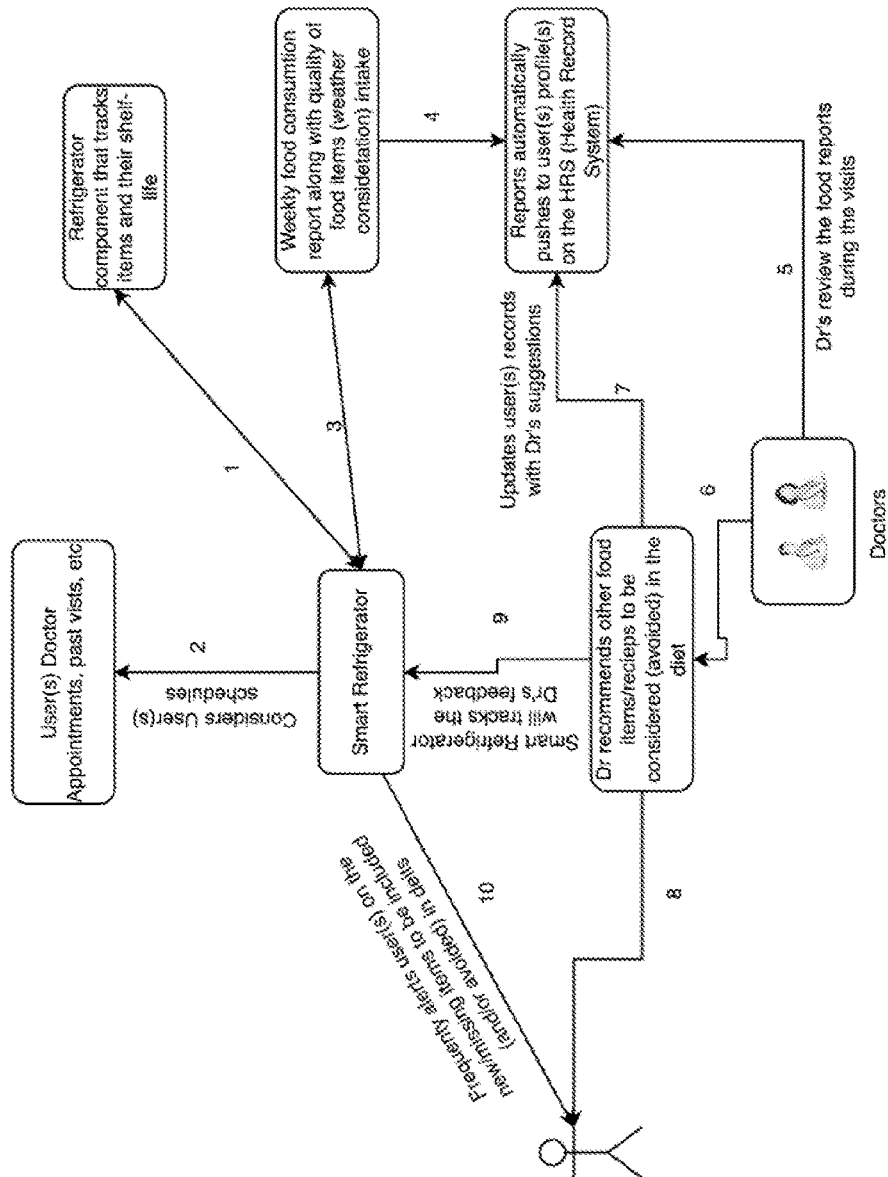


FIG. 5

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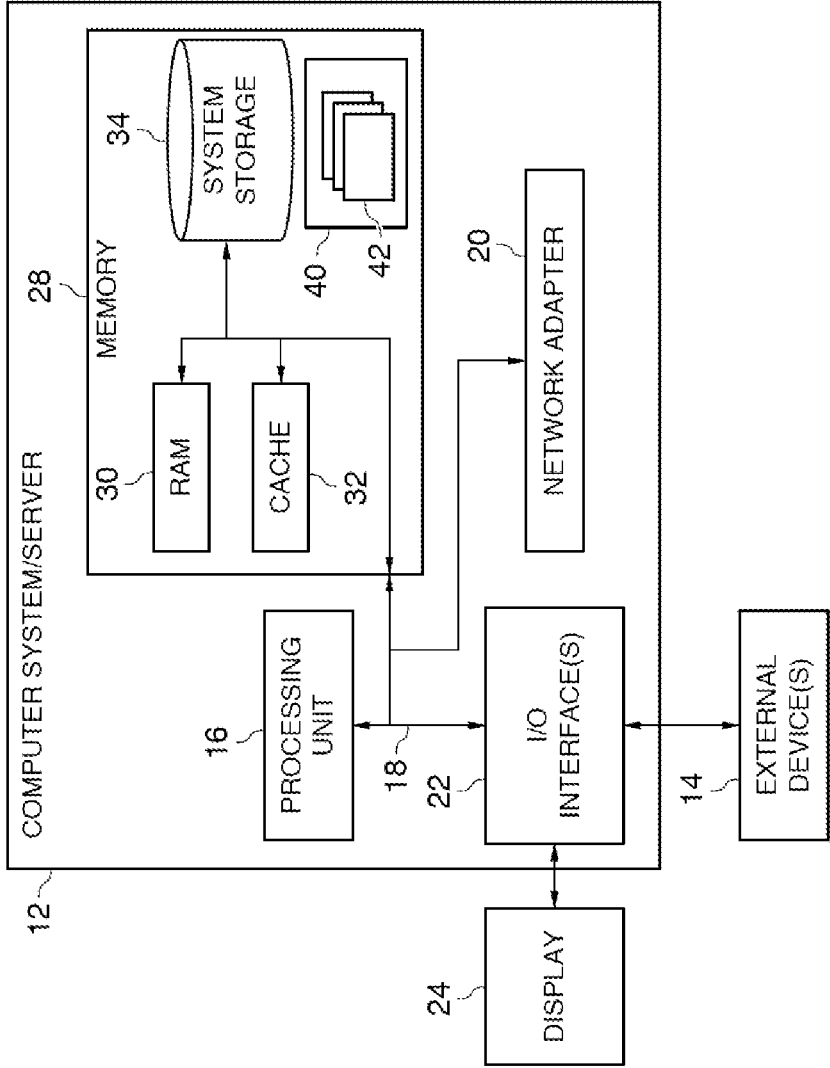


FIG. 6

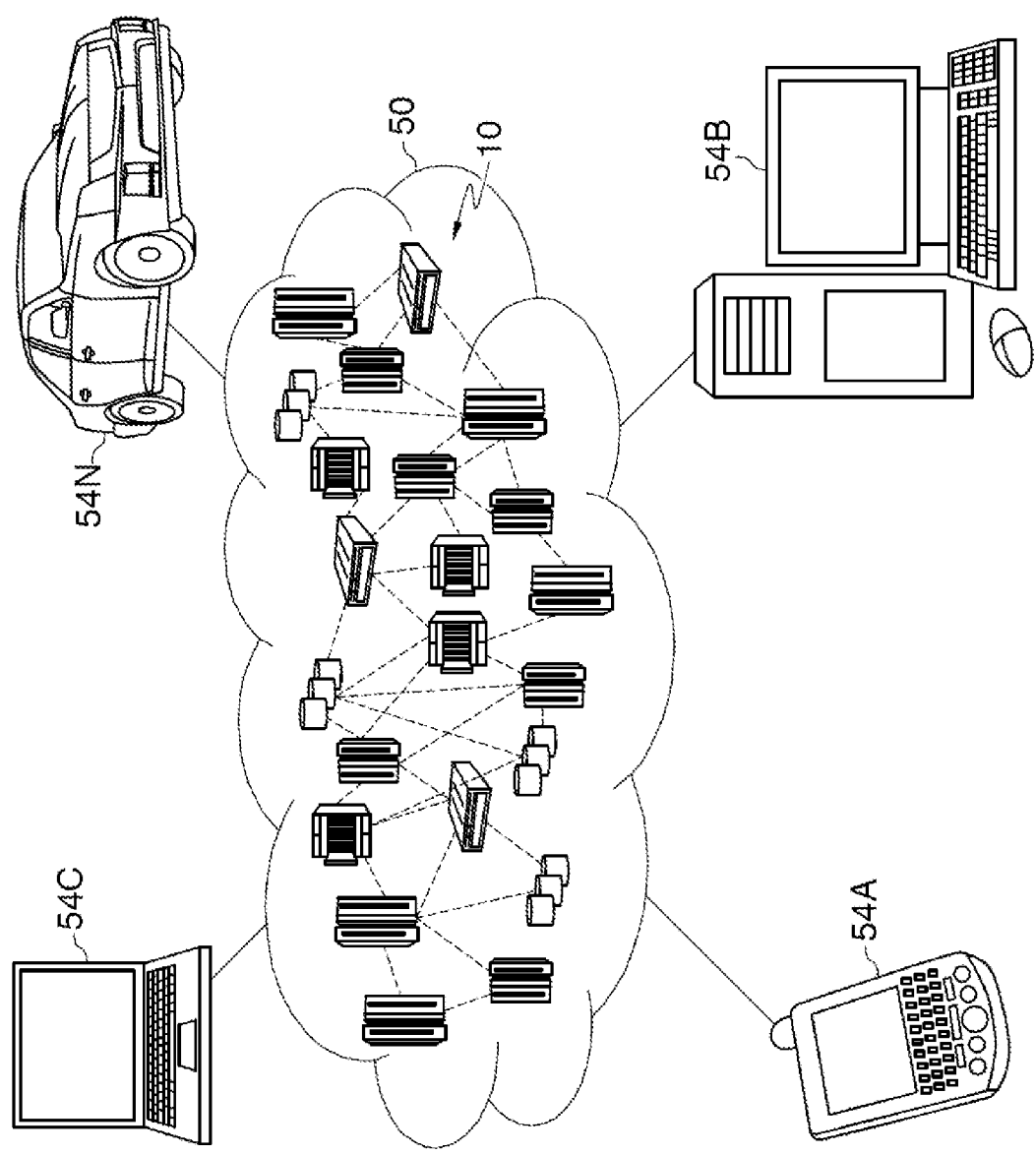


FIG. 7

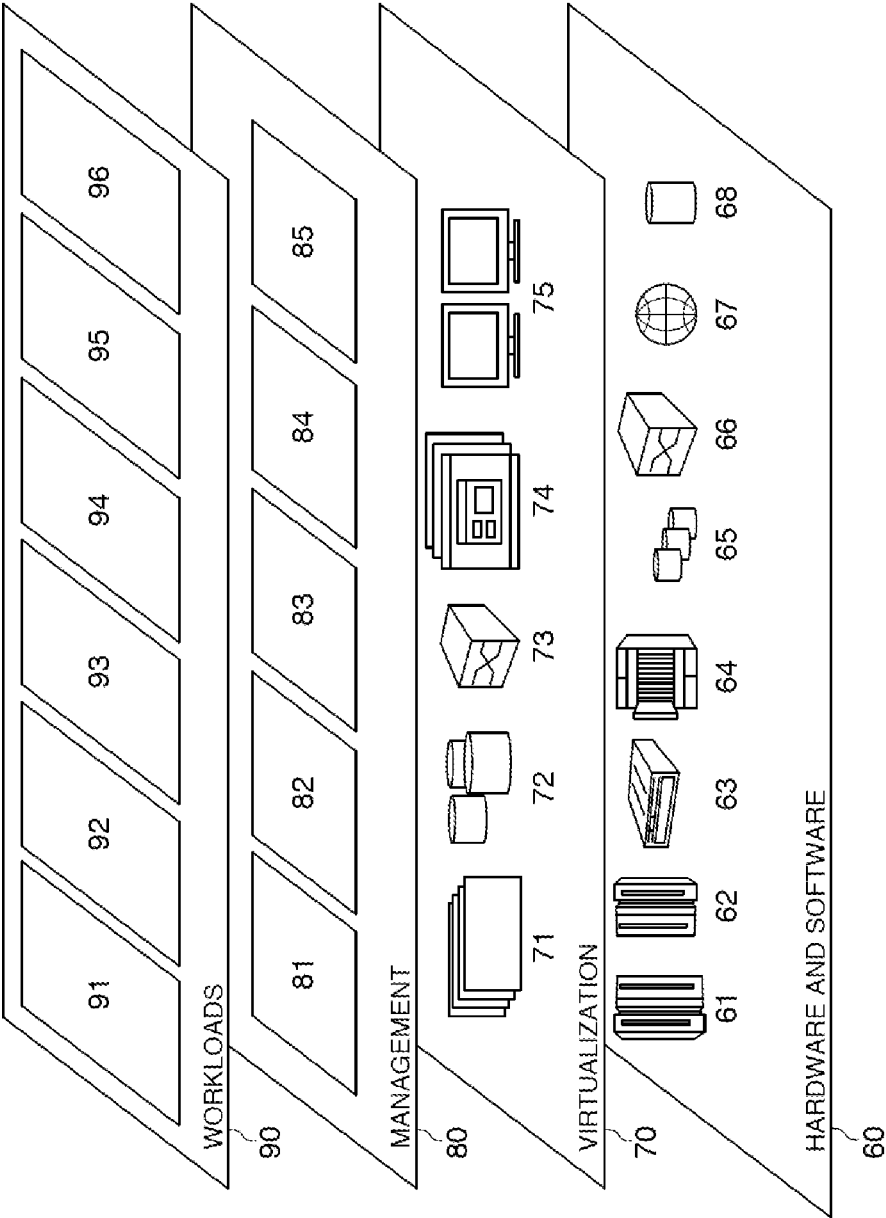


FIG. 8

SMART REFRIGERATOR FOR SMART CONSUMPTION OF FOOD

BACKGROUND

[0001] The exemplary embodiments relate generally to a smart refrigerator (smart fridge), and more particularly to using a smart fridge for smart consumption of food.

[0002] Many people purchase a lot of items and groceries and store them in their refrigerator (fridge). Many people forget to keep track of expiration dates of those items and shelf lives of those groceries. This results in a lot of food wastage. For example, a busy person may forget that their items in their fridge are about to expire in two days. The person may wish to be notified that those items are going to expire in two days, so that they can plan to consume those items prior to expiration and avoid wasting the items.

SUMMARY

[0003] The exemplary embodiments disclose a method, a computer program product, and a computer system for managing a consumption of food. The exemplary embodiments may include collecting data of a user and of one or more food items, extracting one or more features from the collected data, and determining a food consumption recommendation based on the extracted one or more features and one or more models.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0004] The following detailed description, given by way of example and not intended to limit the exemplary embodiments solely thereto, will best be appreciated in conjunction with the accompanying drawings, in which:

[0005] FIG. 1 depicts an exemplary schematic diagram of a smart fridge system **100**, in accordance with the exemplary embodiments.

[0006] FIG. 2 depicts an exemplary flowchart illustrating the operations of a smart fridge manager **134** of the smart fridge system **100** in managing the consumption of food, in accordance with the exemplary embodiments.

[0007] FIG. 3 depicts an exemplary flowchart illustrating the operations of a smart fridge manager **134** of the smart fridge system **100** in determining one or more preferred recipes of a user and recommending one or more recipes to the user, in accordance with the exemplary embodiments.

[0008] FIG. 4 depicts an exemplary flowchart illustrating the operations of a smart fridge manager **134** of the smart fridge system **100** in scheduling the giving away of food, in accordance with the exemplary embodiments.

[0009] FIG. 5 depicts an exemplary flowchart illustrating the operations of a smart fridge manager **134** of the smart fridge system **100** in consulting one or more doctors for feedback and making recommendations to a user based on the feedback from the one or more doctors, in accordance with the exemplary embodiments.

[0010] FIG. 6 depicts an exemplary block diagram depicting the hardware components of the smart fridge system **100** of FIG. 1, in accordance with the exemplary embodiments.

[0011] FIG. 7 depicts a cloud computing environment, in accordance with the exemplary embodiments.

[0012] FIG. 8 depicts abstraction model layers, in accordance with the exemplary embodiments.

[0013] The drawings are not necessarily to scale. The drawings are merely schematic representations, not intended to portray specific parameters of the exemplary embodiments. The drawings are intended to depict only typical exemplary embodiments. In the drawings, like numbering represents like elements.

DETAILED DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0014] Detailed embodiments of the claimed structures and methods are disclosed herein; however, it can be understood that the disclosed embodiments are merely illustrative of the claimed structures and methods that may be embodied in various forms. The exemplary embodiments are only illustrative and may, however, be embodied in many different forms and should not be construed as limited to the exemplary embodiments set forth herein. Rather, these exemplary embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope to be covered by the exemplary embodiments to those skilled in the art. In the description, details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the presented embodiments.

[0015] References in the specification to “one embodiment,” “an embodiment,” “an exemplary embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to implement such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0016] In the interest of not obscuring the presentation of the exemplary embodiments, in the following detailed description, some processing steps or operations that are known in the art may have been combined together for presentation and for illustration purposes and in some instances may have not been described in detail. In other instances, some processing steps or operations that are known in the art may not be described at all. It should be understood that the following description is focused on the distinctive features or elements according to the various exemplary embodiments.

[0017] Many people purchase a lot of items and groceries and store them in their refrigerator (fridge). Many people forget to keep track of expiration dates of those items and shelf lives of those groceries. This results in a lot of food wastage. For example, a busy person may forget that their items in their fridge are about to expire in two days. The person may wish to be notified that those items are going to expire in two days, so that they can plan to consume those items prior to expiration and avoid wasting the items.

[0018] Exemplary embodiments are directed to a method, computer program product, and computer system for managing the consumption of food. In embodiments, machine learning may be used to create models capable of determining one or more food consumption recommendations to a user, while feedback loops may improve upon such models. Moreover, data from user uploads, databases, network **108**,

or the sensors **112** may be used to determine one or more food consumption recommendations to a user.

[0019] Many users may wish for their fridge to keep track of items' shelf lives, expiration dates, compatibility for recipes, etc. in order to reduce food wastage and make food consumption recommendations. In embodiments, food consumption recommendations may pertain to best times to consume food, best recipes to use to consume food, best times to donate food that will not be consumed, etc. For example, a person may anticipate traveling for two weeks on vacation and may be cleaning out the items in their fridge that may expire over those two weeks. The person may wish for their fridge to assist them in identifying items that can be kept until they return two weeks later as well as items that could be donated prior to the two-week vacation. In another example, a commercial chef at a restaurant may wish to know recipes that would use food items that are near expiration, such that the food items will not ultimately be wasted. In general, it will be appreciated that embodiments described herein may relate to aiding in the consumption of food within any environment and for any motivation.

[0020] FIG. 1 depicts the smart fridge system **100**, in accordance with the exemplary embodiments. According to the exemplary embodiments, the smart fridge system **100** may include a smart fridge **110**, a smart device **120**, and a smart fridge server **130**, which may all be interconnected via a network **108**. In embodiments, the smart fridge system **100** may include additional elements such as multiple smart fridges, one or more smart rangiers or smart stoves, etc. While programming and data of the exemplary embodiments may be stored and accessed remotely across several servers via the network **108**, programming and data of the exemplary embodiments may alternatively or additionally be stored locally on as few as one physical computing device or amongst other computing devices than those depicted.

[0021] In the exemplary embodiments, the network **108** may be a communication channel capable of transferring data between connected devices. Accordingly, the components of the smart fridge system **100** may represent network components or network devices interconnected via the network **108**. In the exemplary embodiments, the network **108** may be the Internet, representing a worldwide collection of networks and gateways to support communications between devices connected to the Internet. Moreover, the network **108** may utilize various types of connections such as wired, wireless, fiber optic, etc. which may be implemented as an intranet network, a local area network (LAN), a wide area network (WAN), or a combination thereof. In further embodiments, the network **108** may be a Bluetooth network, a Wi-Fi network, or a combination thereof. In yet further embodiments, the network **108** may be a telecommunications network used to facilitate telephone calls between two or more parties comprising a landline network, a wireless network, a closed network, a satellite network, or a combination thereof. In general, the network **108** may represent any combination of connections and protocols that will support communications between connected devices.

[0022] In the exemplary embodiments, the smart fridge **110** may include one or more sensors **112** and may be one or more smart refrigerator, enterprise server, laptop computer, notebook, tablet computer, netbook computer, personal computer (PC), desktop computer, server, personal digital assistant (PDA), rotary phone, touchtone phone, smart phone, mobile phone, virtual device, thin client,

virtual reality device, augmented reality device, or any other electronic device or computing system capable of receiving and sending data to and from other computing devices. While each smart fridge **110** is shown as a single device, in other embodiments, each smart fridge **110** may be comprised of a cluster or plurality of computing devices, in a modular manner, etc., working together or working independently. The smart fridge may for example comprise multiple bins or sections for various types of food (humidity-controlled crisper bin, temperature-controlled defroster bin, etc.) each with one or more sensors **112**. The smart fridge **110** may additionally comprise one or more mechanisms to move food items from one bin to another, or to eject a food item outside the smart fridge **110**. For example, the smart fridge **110** may comprise one or more mechanical arms (i.e., actuators functioning together) that allow for food items to be picked up and placed elsewhere, for bins to be opened and closed, etc. The smart fridge **110** is described in greater detail as a hardware implementation with reference to FIG. 6, as part of a cloud implementation with reference to FIG. 7, and/or as utilizing functional abstraction layers for processing with reference to FIG. 8.

[0023] In example embodiments, the sensors **112** may comprise a camera, microphone, light sensor, infrared sensor, movement detection sensor, pressure sensor, humidity sensor, thermometer, global positioning system (GPS) sensor, or other sensory hardware equipment. Moreover, the smart fridge **110** may incorporate an array of the one or more sensors **112** such that information can be obtained by the sensors **112** in multiple directions, at different times/intervals, in different mediums/frequencies, and the like. For example, the smart fridge **110** may include three inward-facing cameras located on the inside of the smart fridge **110** door that each record an adjacent sixty-degree viewing angle spanning a total of one-hundred and eighty degrees into the fridge. Moreover, data processing techniques may be implemented such that directional information of visual and audio data can be obtained based on signals received by each of the three sensors **112**, such as trilateration and triangulation. In another example, the smart fridge **110** may include a camera, thermometer, and humidity sensors in each bin or compartment of the smart fridge **110**.

[0024] While the sensors **112** are depicted as integrated with the smart fridge **110**, in embodiments, the sensors **112** may be incorporated within an environment in which the smart fridge system **100** is implemented. For example, the sensors **112** may be one or more thermometers built into a kitchen or living room, a camera built into a facility, etc. Moreover, data processing techniques may be implemented such that directional information of visual and audio data can be obtained based on signals received by each of the sensors **112**, such as trilateration and triangulation. In other embodiments, the sensors **112** may be integrated with other smart devices, e.g., smart phones and laptops, within an environment implementing the smart fridge system **100**. In such embodiments, the sensors **112** may communicate directly with other networks and devices, such as the network **108**. The sensors **112** are described in greater detail as a hardware implementation with reference to FIG. 6, as part of a cloud implementation with reference to FIG. 7, and/or as utilizing functional abstraction layers for processing with reference to FIG. 8.

[0025] In the example embodiment, the smart device **120** includes a smart fridge client **122**, and may be an enterprise

server, a laptop computer, a notebook, a tablet computer, a netbook computer, a personal computer (PC), a desktop computer, a server, a personal digital assistant (PDA), a rotary phone, a touchtone phone, a smart phone, a mobile phone, a virtual device, a thin client, an IoT device, or any other electronic device or computing system capable of receiving and sending data to and from other computing devices. While the smart device **120** is shown as a single device, in other embodiments, the smart device **120** may be comprised of a cluster or plurality of computing devices, in a modular manner, etc., working together or working independently. The smart device **120** is described in greater detail as a hardware implementation with reference to FIG. 6, as part of a cloud implementation with reference to FIG. 7, and/or as utilizing functional abstraction layers for processing with reference to FIG. 8.

[0026] The smart fridge client **122** may act as a client in a client-server relationship. The smart fridge client **122** may also be a software and/or hardware application capable of communicating with and providing a user interface for a user to interact with a server, for example the smart fridge server **130**, via the network **108**. Moreover, in the example embodiment, the smart fridge client **122** may be capable of transferring data from the smart fridge **110** and/or the sensors **112** between the smart device **120** and other devices via the network **108**. In embodiments, the smart fridge client **122** utilizes various wired and wireless connection protocols for data transmission and exchange, including Bluetooth, 2.4GHz and 5GHz internet, near-field communication, Z-Wave, Zigbee, etc. The smart fridge client **122** is described in greater detail with respect to FIG. 2.

[0027] In the exemplary embodiments, the smart fridge server **130** may include one or more smart fridge models **132** and a smart fridge manager **134** and may act as a server in a client-server relationship with the smart fridge client **122**. The smart fridge server **130** may be an enterprise server, a laptop computer, a notebook, a tablet computer, a netbook computer, a PC, a desktop computer, a server, a PDA, a rotary phone, a touchtone phone, a smart phone, a mobile phone, a virtual device, a thin client, an IoT device, or any other electronic device or computing system capable of receiving and sending data to and from other computing devices. While the smart fridge server **130** is shown as a single device, in other embodiments, the smart fridge server **130** may be comprised of a cluster or plurality of computing devices, working together or working independently. The smart fridge server **130** is described in greater detail as a hardware implementation with reference to FIG. 6, as part of a cloud implementation with reference to FIG. 7, and/or as utilizing functional abstraction layers for processing with reference to FIG. 8.

[0028] The smart fridge models **132** may be one or more algorithms modelling a correlation between one or more features detected by the sensors **112** and one or more food consumption recommendations. In the example embodiment, the smart fridge models **132** may be generated using machine learning methods, such as neural networks, deep learning, hierarchical learning, Gaussian Mixture modelling, Hidden Markov modelling, and K-Means, K-Medoids, or Fuzzy C-Means learning, etc., and may model a likelihood of the one or more features being indicative of an accurate or appropriate food consumption recommendation. In embodiments, such features may pertain to items in a smart fridge **110** and may include type, date purchased, expiration

date, shelf life, quantity, temperature, humidity, possible recipes, recipe preparation times, healthiness, etc. In embodiments, such features may additionally pertain to one or more users and may include calendars, medical information, allergy information, dietary restrictions, dietary preferences, etc. The smart fridge models **132** may weight the features based on an effect that the features have on making an appropriate food consumption recommendation. The smart fridge models **132** are described in greater detail with reference to FIG. 2.

[0029] In the exemplary embodiments, the smart fridge manager **134** may be a software and/or hardware program capable of collecting training data, extracting features from the training data, and training one or more models based on the extracted features. The smart fridge manager **134** may additionally be capable of configuring a session and collecting data. The smart fridge manager **134** may further extract features from the collected data and apply one or more models to the extracted features to make one or more food consumption recommendations. Moreover, the smart fridge manager **134** may be further configured for notifying a user of the one or more food consumption recommendations, evaluating whether the one or more recommendations were appropriate, adjusting the one or more models, and facilitating one or more actions based on the one or more recommendations. The smart fridge manager **134** is described in greater detail with reference to FIG. 2.

[0030] FIG. 2 depicts an exemplary flowchart illustrating the operations of a smart fridge manager **134** of the smart fridge system **100** in managing the consumption of food, in accordance with the exemplary embodiments. In exemplary embodiments, the smart fridge manager **134** first implements a training phase in which it trains the smart fridge models **132** using labelled training data from previous interactions between one or more users and one or more smart fridges **110**. The smart fridge manager **134** then moves on to an operational phase in which it applies the trained smart fridge models **132** to current data in order to make one or more appropriate food consumption recommendations.

[0031] The smart fridge manager **134** may collect and/or receive training data (step **204**). In embodiments, the training data may be sourced from previous interactions between one or more users and one or more smart fridges **110**. Training data may include data of one or more user's calendars, medical records, allergy records, dietary restrictions, dietary preferences, locations, number of guests, guests' dietary restrictions, dietary preferences, and audio and video data of the user's smart fridge **110** labelled with one or more food consumption recommendations. For example, training data may include data of a user's allergy to apples and video footage of an apple in the user's smart fridge **110** labelled with recommendation, "Reminder: you are allergic to apples." In a second example, training data may include footage of various food items placed in a smart fridge **110** at various times labelled with recommendation, "Left over pizza will spoil in two days. Perhaps eat pizza for dinner tonight. The bread will soon not be fresh. Perhaps eat the bread with peanut butter as a sandwich for lunch today." In a third example, training data may include footage of various food items placed in a smart fridge **110** at various times along with a user's calendar labelled with recommendation, "It does not appear you will be able to consume the unopened jam and canned beans before their expiration. Perhaps you should donate these items at 10 am tomorrow."

Your calendar shows you are available at this time.” The smart fridge manager **134** may collect training data from microphones and video cameras, which may be built into the user’s smart fridge **110**. The smart fridge manager **134** may retrieve training data via user upload, databases, or the sensors **112**. In embodiments, the smart fridge manager **134** may collect training data via the sensors **112** as one or more microphones built into a kitchen, a camera built into a smart fridge **110** door, etc. The collected training data may pertain to items in a smart fridge **110** and may relate to the type, date purchased, expiration date, shelf life, quantity, temperature, humidity, possible recipes, recipe preparation times, healthiness, etc. of those items. The collected training data may additionally pertain to one or more users and may relate to calendars, medical information, allergy information, dietary restrictions, dietary preferences, etc.

[0032] In embodiments, the smart fridge manager **134** may collect and/or receive training data of one or more appropriate syntaxes for training purposes. For example, training data may include template sentences such as “[Item] will expire on [day],” “Perhaps you should eat [recipe] for [breakfast/lunch/dinner] on [day]” and “It does not appear that you will be able to consume [item] before it expires on [day]. Perhaps you should donate [item] on [day] since your calendar shows you are available.” The smart fridge manager **134** may mirror the syntax of received template sentences when later making one or more food consumption recommendations.

[0033] To further illustrate the operations of the smart fridge manager **134**, reference is now made to an illustrative example where the smart fridge manager **134** collects labelled training data of previous interactions between one or more users and one or more smart fridges **110**. For each user interaction, the smart fridge manager **134** collects data of the contents of a user’s smart fridge **110** and user data from one or more databases labelled with the smart fridge **110**’s appropriate food consumption recommendation to the user. The smart fridge manager **134** additionally collects template sentences for syntax purposes.

[0034] The smart fridge manager **134** may extract one or more features from the collected and/or received training data (step **206**). The extracted features may be extracted from the audio, video, etc. data and/or databases, and may include features pertaining to items in a smart fridge **110** such as type, date purchased, expiration date, shelf life, quantity, temperature, humidity, possible recipes, recipe preparation times, healthiness, etc. In embodiments, such features may additionally pertain to one or more users and may include calendars, medical information, allergy information, dietary restrictions, dietary preferences, etc. In embodiments, the smart fridge manager **134** may use techniques such as feature extraction, natural language processing, named entity recognition, optical character recognition, image processing, video processing, audio processing, pattern/template matching, data comparison, convolutional neural networks, etc. to identify features. For example, the smart fridge manager **134** may extract features such as calendars, medical information, allergy information, dietary restrictions, dietary preferences, etc. directly from one or more databases using feature extraction and optical character recognition. The smart fridge manager **134** may extract features such as shelf life, possible recipes, recipe preparation times, healthiness, etc. directly from the internet via network **108** using image processing, pattern/template

matching, data comparison, etc. The smart fridge manager **134** may additionally extract temperature and humidity directly from thermometers and humidity sensors as the sensors **112** using feature extraction. The smart fridge manager **134** may also extract item types, dates purchased, expiration dates, etc. from collected video and audio footage of one or more sensors **112** (i.e. user placing item in smart fridge **110** or footage of food labels stating expiration dates) using audio processing, video processing, etc. The smart fridge manager **134** may later associate extracted features with the labelled appropriate food consumption recommendations.

[0035] With reference to the previously introduced example where the smart fridge manager **134** collects labelled training data of previous interactions between one or more users and one or more smart fridges **110**, the smart fridge manager **134** extracts features of each item in smart fridges **110** such as type, date purchased, expiration date, shelf life, quantity, temperature, humidity, possible recipes, recipe preparation times, healthiness, etc. as well as features of one or more users such as calendars, medical information, allergy information, dietary restrictions, dietary preferences, etc. from the collected training data with respect to previously received user interactions.

[0036] The smart fridge manager **134** may train one or more smart fridge models **132** based on the extracted features (step **208**). The smart fridge manager **134** may train one or more smart fridge models **132** based on an association of the one or more extracted features with one or more food consumption recommendations. As previously mentioned, such extracted features may include features pertaining to items in a smart fridge **110** such as type, date purchased, expiration date, shelf life, quantity, temperature, humidity, possible recipes, recipe preparation times, healthiness, etc. In embodiments, such features may additionally pertain to one or more users and may include calendars, medical information, allergy information, dietary restrictions, dietary preferences, etc. The one or more smart fridge models **132** may be generated through machine learning techniques such as neural networks, and the smart fridge manager **134** may train the one or more smart fridge models **132** to weight the features such that features shown to have a greater correlation with making an appropriate food consumption recommendation are weighted greater than those features that are not. In embodiments, the smart fridge models **132** may include a model for each user. In other embodiments, the smart fridge manager **134** may simply train one smart fridge model **132** to be later applied to all users. Based on the smart fridge models **132**’s extracted features and weights associated with such extracted features, the smart fridge manager **134** may later determine one or more appropriate food consumption recommendations.

[0037] With reference to the previously introduced example where the smart fridge manager **134** extracts features from the collected training data with respect to previously received user interactions, the smart fridge manager **134** trains a model for each user of the collected training data.

[0038] The smart fridge manager **134** may receive a configuration (step **210**). Having trained the one or more smart fridge models **132**, the smart fridge manager **134** may now apply the one or more smart fridge models **132** to make one or more food consumption recommendations in real time. The smart fridge manager **134** may first, however,

receive a user configuration by receiving a user registration and user preferences. The user registration may be uploaded by a user, i.e., a person using the smart fridge **110** of the smart fridge system **100** or administrator, i.e., a person overseeing the user's usage of the smart fridge system **100** (such as a parent or guardian of the user, employer of the user, etc.) and the configuration may be received by the smart fridge manager **134** via the smart fridge client **122** and the network **108**. Receiving the user registration may involve referencing a user profile via user login credentials, internet protocol (IP) address, media access control (MAC) address, etc., or receiving user input information such as a name, date of birth, gender, address/geographic information, phone number, email address, company name, device serial numbers, one or more smart device **120** types, smart fridge **110** type, sensors **112** types, and the like. Receiving a user registration may also involve receiving or extracting data from databases such as user calendar data, social media data, health data, medical records, etc. Lastly, the smart fridge manager **134** may receive a configuration of the one or more sensors **112**, whether they be fixed to one or more devices (e.g., the smart device **120** or the smart fridge **110**) or fixed within an environment in which the smart fridge system **100** is implemented (e.g., fixed to a stove or fixed to the corner of a kitchen).

[0039] During configuration, the smart fridge manager **134** may further receive user preferences (step **210** continued). User preferences may include preferences for the timing or frequency of notification to the user of one or more recommendations. For example, user preferences may specify that the smart fridge manager **134** is to notify a user of expiring items only one day before their expiration. A different user may specify that the smart fridge manager **134** is to notify them of expiring items two days before their expiration. User preferences may additionally include preferences for the frequency of notification. For example, user preferences may specify that a user be notified on their smart device **120** and their smart fridge **110** once each day of any recommendations, whereas a different user may specify notification only on their smart device **120** but twice each day. User preferences may also specify when the smart fridge manager **134** should facilitate one or more actions based on the one or more recommendations. For example, user preferences may specify that user confirmation is required before events are created in a user's calendar for donating food items. A different user may specify that the smart fridge manager **134** is to schedule donations of food items in their calendar without seeking confirmation from the user.

[0040] With reference to the previously introduced example where the smart fridge manager **134** trains a model for each user of the collected training data, the smart fridge manager **134** receives a user registration via user upload including the user's name, type of smart device **120**, type of smart fridge **110**, types of sensors **112** including a microphone, video camera, thermometer, and humidity sensor, and links to databases containing user calendar data, medical data, allergy data, dietary restrictions, and dietary preferences. The smart fridge manager **134** also receives user preferences via user upload specifying that the user is to be notified of food consumption recommendations once daily via visual feedback on their smart device **120**. The user preferences also specify that the smart fridge manager **134**

must receive confirmation from the user prior to facilitating any actions based on the food consumption recommendations.

[0041] The smart fridge manager **134** may collect data (step **212**). In embodiments, the smart fridge manager **134** may collect data of the user such as user calendar data, social media data, health data, medical data, allergy information, dietary restrictions, dietary preferences, etc. from one or more databases uploaded during configuration. In embodiments, the smart fridge manager **134** may additionally collect data of one or more items in the smart fridge **110** such as type, date purchased, expiration date, shelf life, quantity, temperature, humidity, possible recipes, recipe preparation times, healthiness, etc. from one or more sensors **112**, which may include one or more microphones or video cameras built into the smart fridge **110**, built into a kitchen, etc. In embodiments, the smart fridge manager **134** may continuously collect data from sensors **112** in anticipation of a user opening a door of the smart fridge **110**. In embodiments, the smart fridge manager **134** may begin collecting data from sensors **112** upon one or more trigger, for example a user opening a smart fridge **110** door, opening a smart fridge **110** drawer or compartment, etc. The collected data may relate to data of items in a smart fridge **110** such as type, date purchased, expiration date, shelf life, quantity, temperature, humidity, possible recipes, recipe preparation times, healthiness, etc. In embodiments, collected data may additionally relate to one or more users and may include calendars, medical information, allergy information, dietary restrictions, dietary preferences, etc.

[0042] With reference to the previously introduced example where the smart fridge manager **134** receives a user registration and user preferences, the smart fridge manager **134** collects calendar, medical, allergy, and dietary data of the user from the databases uploaded during configuration. The smart fridge manager **134** additionally collects data of each item in the user's smart fridge **110** from the video camera, thermometer, and humidity sensor inside the smart fridge **110**.

[0043] The smart fridge manager **134** may extract one or more features from the collected and/or received data (step **214**). The smart fridge manager **134** may extract one or more features from the collected and/or received data in the same manner as described with reference to step **206**, however here the features are extracted not from the training data, but rather from the currently collected data.

[0044] With reference to the previously introduced example where the smart fridge manager **134** collects data of the user and the items in the user's smart fridge **110**, the smart fridge manager **134** extracts the below item features in Table 1 and user features in Table 2 from the collected data.

TABLE 1

Extracted Item Features	
leftover pizza	date purchased: Jan. 22, 2021 expiration date: Jan. 28, 2021 quantity: four slices (2 servings) temperature: 35 degrees Fahrenheit humidity: 65%
salad	healthiness: not healthy date purchased: Jan. 24, 2021 expiration date: Feb. 5, 2021 quantity: 3 servings temperature: 35 degrees Fahrenheit

TABLE 1-continued

Extracted Item Features	
chicken	humidity: 65%
	healthiness: very healthy
	date purchased: Jan. 24, 2021
	expiration date: Jan. 30, 2021
	quantity: 2 servings
apple	temperature: 35 degrees Fahrenheit
	humidity: 65%
	healthiness: healthy
	date purchased: Jan. 24, 2021
	expiration date: Feb. 5, 2021
	quantity: 4 apples (4 servings)
	temperature: 35 degrees Fahrenheit
	humidity: 30% (in crisper drawer)
	healthiness: very healthy

TABLE 2

Extracted User Features	
calendar	availability Jan. 25, 2021
medical records	diabetes, high cholesterol
allergy records	allergic to peanut butter
dietary restrictions	lactose intolerant

[0045] The smart fridge manager **134** may apply one or more models to the extracted features to determine one or more food consumption recommendations (step **216**). As previously mentioned, such extracted features may relate to items in a smart fridge **110** such as type, date purchased, expiration date, shelf life, quantity, temperature, humidity, possible recipes, recipe preparation times, healthiness, etc. In embodiments, features may additionally relate to one or more users and may include calendars, medical information, allergy information, dietary restrictions, dietary preferences, etc. and the one or more smart fridge models **132** may be generated through machine learning techniques such as neural networks. In embodiments, the one or more smart fridge models **132** may be trained at initialization and/or through the use of a feedback loop to weight the features such that features shown to have a greater correlation with determining an appropriate food consumption recommendation are weighted greater than those features that are not. Based on the extracted features and weights associated with such extracted features, the smart fridge manager **134** may determine one or more food consumption recommendations such as “Avoid eating [item] as it is bad for your health,” “Consider donating [item] because you may not be able to consume prior to its expiration,” “Consider combining [item] with [item] for a tasty meal,” etc. With reference to FIG. 3-5, applying one or more models to the extracted features may determine food consumption recommendations pertaining to recipe suggestions, scheduling food donations, and medical recommendations.

[0046] With reference to the previously introduced example where the smart fridge manager **134** extracts features from the collected data, the smart fridge manager **134** applies the previously trained model for the user to determine the following appropriate food consumption recommendations: “The leftover pizza is expiring soon and is bad for your health conditions. Perhaps consider donating it during your availability on Jan. 25, 2021 prior to its expi-

ration. Consider combining an apple, chicken, and salad for a tasty and healthy meal that is beneficial for your medical conditions.”

[0047] Upon determining one or more appropriate food consumption recommendations, the smart fridge manager **134** may notify one or more users of the one or more food consumption recommendations (step **218**). In embodiments, the smart fridge manager **134** may notify the user and/or the administrator of the determined recommendations. The smart fridge manager **134** may convey the one or more recommendations to the user and/or administrator in the form of audio, video, text, or any other manner via the smart device **120** and/or the smart fridge **110**. In embodiments, the smart fridge manager **134** may notify the user and/or administrator of multiple food consumption recommendations at once. In other embodiments, the smart fridge manager **134** may notify the user and/or administrator of one recommendation at a time in a queue, and the smart fridge manager **134** may await feedback from the user and/or administrator indicative of whether one or more of the recommendations are appropriate. As discussed with reference to configuration, the smart fridge manager **134** may notify the user and/or administrator of one or more recommendations according to the user preferences of configuration. In embodiments, the smart fridge manager **134** may be configured for receiving user and/or administrator input acknowledging, dismissing, and/or affirming that one or more recommendations are appropriate and/or that one or more actions should be facilitated based on the one or more recommendations. With reference to FIG. 3-5, one or more users may be notified of food consumption recommendations pertaining to recipe suggestions, scheduling food donations, and medical recommendations.

[0048] With reference to the previously introduced example where the smart fridge manager **134** recommends, “The leftover pizza is expiring soon and is bad for your health conditions. Perhaps consider donating it during your availability on Jan. 25, 2021 prior to its expiration. Consider combining an apple, chicken, and salad for a tasty and healthy meal that is beneficial for your medical conditions,” the smart fridge manager **134** notifies the user of the recommendations on their smart device via visual feedback according to the user preferences.

[0049] The smart fridge manager **134** may evaluate and modify the smart fridge models **132** (step **220**). In the example embodiment, the smart fridge manager **134** may verify whether the one or more recommendations were appropriate and/or helpful in order to provide a feedback loop for modifying the smart fridge models **132**. In embodiments, the feedback loop may simply provide a means for the user and/or administrator to indicate whether the one or more food consumption recommendations were appropriate and/or helpful to the user. The feedback loop indication may be triggered via a toggle switch, button, slider, etc. that may be selected by the user and/or administrator manually by hand using a button/touchscreen/etc., by voice, and the like. Based on the smart fridge manager **134** appropriately or inappropriately making one or more food consumption recommendations, the smart fridge manager **134** may modify the smart fridge models **132**. In other embodiments, the smart fridge manager **134** may infer or deduce whether the recommendation was helpful to the user. For example, the smart fridge manager **134** may interpret user dialogue via natural language processing to determine whether the

reply was helpful to the user. For example, if the user says, “No way! That doesn’t help at all” or other expressions indicative of a user’s dissatisfaction, the smart fridge manager **134** may infer that the recommendation was inappropriate and modify the smart fridge models **132** accordingly. Based on feedback received in the above or any other manners, the smart fridge manager **134** may then modify the smart fridge models **132** to more accurately make food consumption recommendations.

[0050] With reference to the previously introduced example where the smart fridge manager **134** notifies the user: “The leftover pizza is expiring soon and is bad for your health conditions. Perhaps consider donating it during your availability on Jan. 25, 2021 prior to its expiration. Consider combining an apple, chicken, and salad for a tasty and healthy meal that is beneficial for your medical conditions,” on their smart device via visual feedback according to the user preferences, the user says, “Thank you, that helps. Let’s do that.” and the smart fridge manager **134** modifies the smart fridge models **132** accordingly.

[0051] The smart fridge manager **134** may facilitate one or more actions based on the one or more food consumption recommendations (step **222**). As previously discussed, in embodiments, the smart fridge manager **134** may require user confirmation, approval, feedback, etc. (i.e., from step **220**) in order to facilitate one or more actions. In other embodiments, the smart fridge manager **134** may simply facilitate one or more actions without requiring user confirmation. In embodiments, and with reference to FIG. 4, the smart fridge manager **134** may facilitate a scheduling of an appointment to donate food. For example, if the smart fridge manager **134** determines that a user should donate a food item at a certain time and/or day based on the user’s calendar or schedule, the smart fridge manager **134** may schedule an event in the user’s calendar and/or contact (i.e., via phone call, text message, email message, scheduling website, etc.) a food donation site to schedule an appointment. In embodiments, the smart fridge manager **134** may facilitate the preparation of one or more food items for consumption. For example, if a user confirms that they would like to eat an item soon for lunch, the smart fridge manager **134** may begin defrosting the item (i.e., shift item to a certain section of the smart fridge **110** with warmer temperature or eject food item outside the smart fridge **110** to a warmer temperature). In embodiments, and with reference to FIG. 5, the smart fridge manager **134** may contact one or more doctors, nurses, health professionals, etc. regarding the user’s diet and/or health. For example, the smart fridge manager **134** may notify a user’s doctor if the doctor asked the user to limit their sugar intake to a specified level but the user is consuming drastically more sugar than the specified level. In embodiments, the smart fridge manager **134** may facilitate one or more other actions based on the determined one or more recommendations.

[0052] With reference to the previously introduced example where the user says, “Thank you, that helps. Let’s do that.” and the smart fridge manager **134** modifies the smart fridge models **132** accordingly, the smart fridge manager **134** schedules an appointment with a local homeless shelter for donation of the leftover pizza on Jan. 25, 2021, and updates the user’s calendar to reflect this appointment. The smart fridge manager **134** additionally begins warming the temperature of the apple, chicken, and salad in preparation for the user’s consumption.

[0053] FIG. 3 depicts an exemplary flowchart illustrating the operations of a smart fridge manager **134** of the smart fridge system **100** in determining one or more preferred recipes of a user and recommending one or more recipes to the user, in accordance with the exemplary embodiments.

[0054] FIG. 4 depicts an exemplary flowchart illustrating the operations of a smart fridge manager **134** of the smart fridge system **100** in scheduling the giving away of food, in accordance with the exemplary embodiments.

[0055] FIG. 5 depicts an exemplary flowchart illustrating the operations of a smart fridge manager **134** of the smart fridge system **100** in consulting one or more doctors for feedback and making recommendations to a user based on the feedback from the one or more doctors, in accordance with the exemplary embodiments.

[0056] FIG. 6 depicts a block diagram of devices within the smart fridge system **100** of FIG. 1, in accordance with the exemplary embodiments. It should be appreciated that FIG. 6 provides only an illustration of one implementation and does not imply any limitations with regard to the environments in which different embodiments may be implemented. Many modifications to the depicted environment may be made.

[0057] Devices used herein may include one or more processors **02**, one or more computer-readable RAMs **04**, one or more computer-readable ROMs **06**, one or more computer readable storage media **08**, device drivers **12**, read/write drive or interface **14**, network adapter or interface **16**, all interconnected over a communications fabric **18**. Communications fabric **18** may be implemented with any architecture designed for passing data and/or control information between processors (such as microprocessors, communications and network processors, etc.), system memory, peripheral devices, and any other hardware components within a system.

[0058] One or more operating systems **10**, and one or more application programs **11** are stored on one or more of the computer readable storage media **08** for execution by one or more of the processors **02** via one or more of the respective RAMs **04** (which typically include cache memory). In the illustrated embodiment, each of the computer readable storage media **08** may be a magnetic disk storage device of an internal hard drive, CD-ROM, DVD, memory stick, magnetic tape, magnetic disk, optical disk, a semiconductor storage device such as RAM, ROM, EPROM, flash memory or any other computer-readable tangible storage device that can store a computer program and digital information.

[0059] Devices used herein may also include a RAY drive or interface **14** to read from and write to one or more portable computer readable storage media **26**. Application programs **11** on said devices may be stored on one or more of the portable computer readable storage media **26**, read via the respective RAY drive or interface **14** and loaded into the respective computer readable storage media **08**.

[0060] Devices used herein may also include a network adapter or interface **16**, such as a TCP/IP adapter card or wireless communication adapter (such as a 4G wireless communication adapter using OFDMA technology). Application programs **11** on said computing devices may be downloaded to the computing device from an external computer or external storage device via a network (for example, the Internet, a local area network or other wide area network or wireless network) and network adapter or interface **16**. From the network adapter or interface **16**, the

programs may be loaded onto computer readable storage media **08**. The network may comprise copper wires, optical fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers.

[0061] Devices used herein may also include a display screen **20**, a keyboard or keypad **22**, and a computer mouse or touchpad **24**. Device drivers **12** interface to display screen **20** for imaging, to keyboard or keypad **22**, to computer mouse or touchpad **24**, and/or to display screen **20** for pressure sensing of alphanumeric character entry and user selections. The device drivers **12**, R/W drive or interface **14** and network adapter or interface **16** may comprise hardware and software (stored on computer readable storage media **08** and/or ROM **06**).

[0062] The programs described herein are identified based upon the application for which they are implemented in a specific one of the exemplary embodiments. However, it should be appreciated that any particular program nomenclature herein is used merely for convenience, and thus the exemplary embodiments should not be limited to use solely in any specific application identified and/or implied by such nomenclature.

[0063] Based on the foregoing, a computer system, method, and computer program product have been disclosed. However, numerous modifications and substitutions can be made without deviating from the scope of the exemplary embodiments. Therefore, the exemplary embodiments have been disclosed by way of example and not limitation.

[0064] It is to be understood that although this disclosure includes a detailed description on cloud computing, implementation of the teachings recited herein are not limited to a cloud computing environment. Rather, the exemplary embodiments are capable of being implemented in conjunction with any other type of computing environment now known or later developed.

[0065] Cloud computing is a model of service delivery for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, network bandwidth, servers, processing, memory, storage, applications, virtual machines, and services) that can be rapidly provisioned and released with minimal management effort or interaction with a provider of the service. This cloud model may include at least five characteristics, at least three service models, and at least four deployment models.

[0066] Characteristics are as follows:

[0067] On-demand self-service: a cloud consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with the service's provider.

[0068] Broad network access: capabilities are available over a network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, laptops, and PDAs).

[0069] Resource pooling: the provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to demand. There is a sense of location independence in that the consumer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or data center).

[0070] Rapid elasticity: capabilities can be rapidly and elastically provisioned, in some cases automatically, to quickly scale out and rapidly released to quickly scale in. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be purchased in any quantity at any time.

[0071] Measured service: cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

[0072] Service Models are as follows:

[0073] Software as a Service (SaaS): the capability provided to the consumer is to use the provider's applications running on a cloud infrastructure. The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based e-mail). The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.

[0074] Platform as a Service (PaaS): the capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages and tools supported by the provider. The consumer does not manage or control the underlying cloud infrastructure including networks, servers, operating systems, or storage, but has control over the deployed applications and possibly application hosting environment configurations.

[0075] Infrastructure as a Service (IaaS): the capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications. The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of select networking components (e.g., host firewalls).

[0076] Deployment Models are as follows:

[0077] Private cloud: the cloud infrastructure is operated solely for an organization. It may be managed by the organization or a third party and may exist on-premises or off-premises.

[0078] Community cloud: the cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be managed by the organizations or a third party and may exist on-premises or off-premises.

[0079] Public cloud: the cloud infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services.

[0080] Hybrid cloud: the cloud infrastructure is a composition of two or more clouds (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load-balancing between clouds).

[0081] A cloud computing environment is service oriented with a focus on statelessness, low coupling, modularity, and semantic interoperability. At the heart of cloud computing is an infrastructure that includes a network of interconnected nodes.

[0082] Referring now to FIG. 7, illustrative cloud computing environment 50 is depicted. As shown, cloud computing environment 50 includes one or more cloud computing nodes 40 with which local computing devices used by cloud consumers, such as, for example, personal digital assistant (PDA) or cellular telephone 54A, desktop computer 54B, laptop computer 54C, and/or automobile computer system 54N may communicate. Nodes 40 may communicate with one another. They may be grouped (not shown) physically or virtually, in one or more networks, such as Private, Community, Public, or Hybrid clouds as described hereinabove, or a combination thereof. This allows cloud computing environment 50 to offer infrastructure, platforms and/or software as services for which a cloud consumer does not need to maintain resources on a local computing device. It is understood that the types of computing devices 54A-N shown in FIG. 7 are intended to be illustrative only and that computing nodes 40 and cloud computing environment 50 can communicate with any type of computerized device over any type of network and/or network addressable connection (e.g., using a web browser).

[0083] Referring now to FIG. 8, a set of functional abstraction layers provided by cloud computing environment 50 (FIG. 7) is shown. It should be understood in advance that the components, layers, and functions shown in FIG. 8 are intended to be illustrative only and the exemplary embodiments are not limited thereto. As depicted, the following layers and corresponding functions are provided:

[0084] Hardware and software layer 60 includes hardware and software components. Examples of hardware components include: mainframes 61; RISC (Reduced Instruction Set Computer) architecture based servers 62; servers 63; blade servers 64; storage devices 65; and networks and networking components 66. In some embodiments, software components include network application server software 67 and database software 68.

[0085] Virtualization layer 70 provides an abstraction layer from which the following examples of virtual entities may be provided: virtual servers 71; virtual storage 72; virtual networks 73, including virtual private networks; virtual applications and operating systems 74; and virtual clients 75.

[0086] In one example, management layer 80 may provide the functions described below. Resource provisioning 81 provides dynamic procurement of computing resources and other resources that are utilized to perform tasks within the cloud computing environment. Metering and Pricing 82 provide cost tracking as resources are utilized within the cloud computing environment, and billing or invoicing for consumption of these resources. In one example, these resources may include application software licenses. Security provides identity verification for cloud consumers and tasks, as well as protection for data and other resources. User portal 83 provides access to the cloud computing environment for consumers and system administrators. Service level management 84 provides cloud computing resource allocation and management such that required service levels are met. Service Level Agreement (SLA) planning and fulfillment 85 provide pre-arrangement for, and procurement of,

cloud computing resources for which a future requirement is anticipated in accordance with an SLA.

[0087] Workloads layer 90 provides examples of functionality for which the cloud computing environment may be utilized. Examples of workloads and functions which may be provided from this layer include: mapping and navigation 91; software development and lifecycle management 92; virtual classroom education delivery 93; data analytics processing 94; transaction processing 95; and food consumption recommendation 96.

[0088] The exemplary embodiments may be a system, a method, and/or a computer program product at any possible technical detail level of integration. The computer program product may include a computer readable storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out aspects of the exemplary embodiments.

[0089] The computer readable storage medium can be a tangible device that can retain and store instructions for use by an instruction execution device. The computer readable storage medium may be, for example, but is not limited to, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of the foregoing. A non-exhaustive list of more specific examples of the computer readable storage medium includes the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device such as punch-cards or raised structures in a groove having instructions recorded thereon, and any suitable combination of the foregoing. A computer readable storage medium, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

[0090] Computer readable program instructions described herein can be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network and/or a wireless network. The network may comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device.

[0091] Computer readable program instructions for carrying out operations of the exemplary embodiments may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, configuration data for integrated circuitry, or either source code or object code written in any combination of one

or more programming languages, including an object oriented programming language such as Smalltalk, C++, or the like, and procedural programming languages, such as the “C” programming language or similar programming languages. The computer readable program instructions may execute entirely on the user’s computer, partly on the user’s computer, as a stand-alone software package, partly on the user’s computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user’s computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects of the exemplary embodiments.

[0092] Aspects of the exemplary embodiments are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products according to the exemplary embodiments. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer readable program instructions.

[0093] These computer readable program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart and/or block diagram block or blocks.

[0094] The computer readable program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0095] The flowchart and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various exemplary embodiments. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function

(s). In some alternative implementations, the functions noted in the blocks may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

What is claimed is:

1. A computer-implemented method for managing a consumption of food, the method comprising:

collecting data of a user and of one or more food items; extracting one or more features from the collected data; and

determining a food consumption recommendation based on the extracted one or more features and one or more models.

2. The method of claim 1, further comprising:

notifying the user of the determined food consumption recommendation.

3. The method of claim 2, further comprising:

receiving a confirmation of the determined food consumption recommendation from the user;

adjusting the one or more models based on the received confirmation; and

facilitating one or more actions based on the food consumption recommendation.

4. The method of claim 3, wherein the one or more actions comprise scheduling a time to donate food, defrosting one or more of the food items, or notifying one or more medical professionals.

5. The method of claim 1, wherein the one or more models correlate the one or more features with the likelihood of determining an appropriate food consumption recommendation.

6. The method of claim 1, further comprising:

collecting training data;

extracting training features from the training data; and

training one or more models based on the extracted training features.

7. The method of claim 1, wherein the one or more features include item types, dates purchased of items, expiration dates of items, shelf lives of items, quantities of items, temperatures of items, humidities of items, possible recipes of items, recipe preparation times of items, healthiness of items, user calendars, user schedules, user medical information, user allergy information, user dietary restrictions, and user dietary preferences.

8. A computer program product for managing a consumption of food, the computer program product comprising:

one or more non-transitory computer-readable storage media and program instructions stored on the one or more non-transitory computer-readable storage media capable of performing a method, the method comprising:

collecting data of a user and of one or more food items; extracting one or more features from the collected data; and

determining a food consumption recommendation based on the extracted one or more features and one or more models.

9. The computer program product of claim 8, further comprising:

notifying the user of the determined food consumption recommendation.

10. The computer program product of claim 9, further comprising:

receiving a confirmation of the determined food consumption recommendation from the user;

adjusting the one or more models based on the received confirmation; and

facilitating one or more actions based on the food consumption recommendation.

11. The computer program product of claim 10, wherein the one or more actions comprise scheduling a time to donate food, defrosting one or more of the food items, or notifying one or more medical professionals.

12. The computer program product of claim 8, wherein the one or more models correlate the one or more features with the likelihood of determining an appropriate food consumption recommendation.

13. The computer program product of claim 8, further comprising:

collecting training data;

extracting training features from the training data; and

training one or more models based on the extracted training features.

14. The computer program product of claim 8, wherein the one or more features include item types, dates purchased of items, expiration dates of items, shelf lives of items, quantities of items, temperatures of items, humidities of items, possible recipes of items, recipe preparation times of items, healthiness of items, user calendars, user schedules, user medical information, user allergy information, user dietary restrictions, and user dietary preferences.

15. A computer system for managing a consumption of food, the computer system comprising:

one or more computer processors, one or more computer-readable storage media, and program instructions stored on the one or more of the computer-readable storage media for execution by at least one of the one or more processors capable of performing a method, the method comprising:

collecting data of a user and of one or more food items; extracting one or more features from the collected data; and

determining a food consumption recommendation based on the extracted one or more features and one or more models.

16. The computer system of claim 15, further comprising: notifying the user of the determined food consumption recommendation.

17. The computer system of claim 16, further comprising: receiving a confirmation of the determined food consumption recommendation from the user;

adjusting the one or more models based on the received confirmation; and

facilitating one or more actions based on the food consumption recommendation.

18. The computer system of claim 17, wherein the one or more actions comprise scheduling a time to donate food, defrosting one or more of the food items, or notifying one or more medical professionals.

19. The computer system of claim 15, wherein the one or more models correlate the one or more features with the likelihood of determining an appropriate food consumption recommendation.

20. The computer system of claim 15, further comprising: collecting training data;

extracting training features from the training data; and

training one or more models based on the extracted training features.

* * * * *